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Cambridge IGCSE Combined and Co-ordinated Sciences

B2 Movement into and out of Cells

Comprehensive Exam Preparation Guide

B2 Movement into and out of Cells — Cambridge IGCSE Biology Study Guide

Chapter Overview

In this chapter you will:

- Learn about diffusion
- Investigate factors that affect the rate of diffusion
- Find out how osmosis is a special kind of diffusion, involving water
- Investigate osmosis, using plant tissues
- Learn about active transport
- Learn about water potential
- Explain how osmosis affects plant cells
- Explain how active transport happens

All learners study some content in this chapter. The big idea is that substances move into and out of cells in three ways: **diffusion** (any particles, down a concentration gradient), **osmosis** (water only, through a partially permeable membrane) and **active transport** (against a concentration gradient, using energy from respiration).

Important Terms

Term	Definition
Particles	(In this context) the smallest pieces of which a substance is made; particles can be molecules, atoms or ions
Diffusion	The net movement of particles from a region of their higher concentration to a region of their lower concentration (i.e. down a concentration gradient), as a result of their random movement
Net movement	Overall or average movement
Concentration gradient	An imaginary 'slope' from a high concentration to a low concentration
Random movement	Movement in any direction, purely by chance; particles travel in straight lines and change direction when they collide with other particles
Kinetic energy	The energy of movement; the energy for diffusion and osmosis comes from the kinetic energy of the moving particles, not from respiration
Rate of diffusion	How fast particles diffuse; it is faster at a higher temperature, with a larger surface area, a steeper concentration gradient and a shorter diffusion distance
Diffusion distance	The distance particles have to travel, e.g. the thickness of a membrane; the shorter the distance, the faster the diffusion
Surface area to volume ratio	The surface area of an object divided by its volume; cutting a cube into smaller cubes increases this ratio, so substances diffuse in faster
Independent variable	The variable that you change in an experiment

Dependent variable	The variable that you measure as you collect your results
Standardised variable	A variable that you keep the same throughout an experiment so that it cannot affect the results (also called a control variable)
Solvent	The liquid in which a substance dissolves; in the solutions in and around cells, the solvent is water
Solute	A substance that is dissolved in a solvent, e.g. sugar dissolved in water
Partially permeable membrane	A membrane that lets some molecules through (e.g. small water molecules) but not others (e.g. much larger sugar molecules); cell membranes are partially permeable
Fully permeable	Lets all molecules pass through; the cell wall of a plant cell is fully permeable
Osmosis	The diffusion of water molecules through a partially permeable membrane
Osmosis (in terms of water potential)	The net movement of water molecules from a region of higher water potential (dilute solution) to a region of lower water potential (concentrated solution) through a partially permeable membrane
Water potential	A way of describing how much water a solution contains compared with dissolved solute; a dilute solution has a high water potential, a concentrated solution has a low water potential, and pure water has the highest water potential of all
High water potential	An area where there are a lot of water molecules, that is, a dilute solution
Low water potential	An area where there are not many water molecules, that is, a concentrated solution
Water potential gradient	A difference in water potential between two areas; water moves down it, from a high to a low water potential
Turgid	A description of a plant cell that is tight and firm
Turgor pressure	The pressure of the water pushing outwards on a plant cell wall
Flaccid	A description of a plant cell that has lost its turgor pressure and is soft
Plasmolysed	A description of a cell in which the cell membrane tears away from the cell wall
Plasmolysis	What happens when a plant cell loses so much water by osmosis that its cell membrane tears away from the cell wall; it can kill the cell
Wilting	The drooping of a plant that happens when its cells become flaccid and the plant loses its firmness
Active transport	The movement of molecules or ions through a cell membrane from a region of lower concentration to a region of higher concentration (i.e. against a concentration gradient) using energy from respiration
Carrier proteins (protein carriers)	Protein molecules in cell membranes that can use energy to change shape and move ions or molecules into or out of a cell
Aerobic respiration	The process in mitochondria that releases energy for the cell to use; some of this energy is used for active transport
Root hair cell	A cell on the surface of a plant root that takes in water and mineral ions from the soil; it uses active transport to take in nitrate ions

Dialysis	A treatment for kidney failure in which unwanted substances diffuse out of the blood, through a very thin membrane, into dialysis fluid that then flows away
Indicator	A dye that changes colour depending on pH; cresol red is red-purple in alkaline conditions and turns yellow when acid is added
Anomalous result	A result that does not fit the pattern or trend shown by the other results

B2.01 Diffusion

Before You Start: Particles in Solids, Liquids and Gases

The particle theory helps us to explain how matter behaves. Diffusion depends on particles being able to move, so it is worth checking that you remember how particles behave in each state of matter.

Q: Here are six statements about particles. Some of them describe solids, some describe liquids and some describe gases. Which are which?

A: Gases: statements 1 and 4. Solids: statements 2 and 6. Liquids: statements 3 and 5.

Statement	State of matter
1. The particles are far apart and rarely collide with each other.	Gas
2. The particles vibrate on the spot.	Solid
3. The particles are close to one another but can move around.	Liquid
4. The particles move freely in all directions.	Gas
5. The particles stay in contact and slide past one another as they move.	Liquid
6. The particles are in fixed positions, close to one another.	Solid

Because particles in gases and liquids can move around freely, diffusion happens in gases, solutions and mixtures of liquids. In a solid the particles only vibrate in fixed positions, so they cannot spread out.

Science in Context B2.01: Filtering Blood

We each have two kidneys. Their job is to remove unwanted substances from the blood. The kidneys produce a solution of the unwanted substances, which is removed from the body as **urine**. Sometimes kidneys stop working. Then the unwanted substances build up in the blood, and some of them can be quite dangerous, so another way must be found to remove them.

Figure B2.01 is a photograph of a boy in a hospital bed attached to a **dialysis machine**. This is how dialysis works:

1. Blood from a vein in his arm flows into the machine.
2. Inside the machine, a liquid called **dialysis fluid** flows past his blood. There is a very thin membrane between the blood and the dialysis fluid.
3. The unwanted substances in his blood diffuse through the membrane into the dialysis fluid (from a high concentration in the blood to a low concentration in the fluid).
4. The dialysis fluid, now carrying the unwanted substances, flows away.
5. His cleaned blood flows back into his body.

It takes several hours to remove all the unwanted substances from his blood, and he has to do this several times a week. Dialysis is a real-life use of **diffusion through a membrane**.

Q: (1) If the boy could have a kidney transplant, he would not have to spend time attached to a dialysis machine. Why do you think not everyone who needs a kidney transplant can have one?

A: There are not enough donated kidneys for everyone who needs one, so there are long waiting lists. A donated kidney must also be a close match to the patient's tissues, otherwise the patient's immune system will reject it, so a suitable match may not be available. Some patients may also be too ill to survive a major operation.

Q: (2) The dialysis fluid flows constantly through the machine. Can you suggest why this works better than just having stationary fluid in the machine?

A: Flowing fluid carries the unwanted substances away as soon as they diffuse into it, and fresh fluid containing none of these substances keeps arriving. This keeps the concentration of unwanted substances in the fluid low, so a steep concentration gradient is maintained between the blood and the fluid, and diffusion out of the blood continues quickly. In stationary fluid the unwanted substances would build up until their concentration was the same as in the blood. There would then be no concentration gradient, and net diffusion would stop.

What Is Diffusion?

Everything — including living cells — is made of atoms, molecules and ions. In this chapter we call all of these **particles**. Particles are always moving. When they can move freely, particles tend to spread themselves out evenly. This happens with gases, solutions and mixtures of liquids, and it is called **diffusion**.

Definition: Diffusion is the net movement of particles from a region of their higher concentration to a region of their lower concentration (i.e. down a concentration gradient), as a result of their random movement.

The Rotten Egg Example (Figure B2.02)

Imagine a rotten egg full of very smelly **hydrogen sulfide** gas in one corner of a room.

- **Figure B2.02a:** at the start there is a high concentration of hydrogen sulfide molecules in one corner (near the egg) and none in the rest of the room.
- **Figure B2.02b:** given time, the molecules spread evenly through all the space available.
- Soon you cannot tell where the smell first came from, because the whole room smells of rotten egg.

Random Movement and Net Movement

- Diffusion happens because the molecules move **randomly**. They do not purposefully move from one place to another.
- Each molecule just moves by chance, changing direction when it bumps into another molecule. Some molecules might even go back into the rotten egg.
- **Net movement** means overall or average movement. Some molecules go from the egg into the room and a few go from the room back into the egg, but overall most of them go from the egg into the room.
- The energy for this movement is the **kinetic energy** of the particles themselves. Cells do not have to supply any energy from respiration for diffusion.

Three Conditions for Diffusion

The definition tells us that diffusion needs: (1) particles that can **move**, (2) a **concentration gradient**, and (3) **random movement** of the particles. Question B2.01 asks you to explain why each one is needed.

Concentration Gradients (Figure B2.03)

A place where a lot of molecules are close together has a **high concentration** of molecules. In the example, the rotten egg had the highest concentration of hydrogen sulfide molecules and the rest of the room had a low concentration.

Figure B2.03 uses a slope to picture this. Lots of particles are crowded at the top of the slope (high concentration of particles), there are only a few at the bottom (low concentration of particles), and an arrow points down the slope. We can imagine particles moving 'down the slope', from where there is a high concentration to where there is a low concentration. Another word for 'slope' is 'gradient', so we call this a **concentration gradient**. The overall result of diffusion is that particles move 'down' their concentration gradient.

Activity B2.01: Modelling Diffusion

Aim: To model diffusion, using people to represent the particles of a gas.

You will need: a fairly big, empty space to move around in, such as a playground or a large hall.

Method summary:

1. Work as a whole class. Each person represents a particle in a gas.
2. Start with most people in one corner of the space. Do not crowd too closely, because gas particles do not touch one another. Face any way you like.
3. Each person moves in a straight line in the direction they are facing.
4. Keep going until you hit something (another 'particle' or the wall of the 'container'). When you collide, change direction and keep going in a new straight line.
5. Continue for a few minutes, then stop and look around.

Safety: Real gas particles move very quickly, but do not move too fast, or there might be uncomfortable collisions.

Expected result: by the end, the 'particles' are spread out fairly evenly over the whole space.

Q: (1) What has happened to the 'particles'? Are they still concentrated in one corner?

A: No. The 'particles' have spread out across the whole space and are now fairly evenly spread, instead of being crowded into one corner.

Q: (2) Try to explain why this has happened.

A: Each person moved in a straight line and changed direction at random every time they collided with someone or something. Nobody was trying to leave the corner, but because most of the people started in the corner, by chance far more people moved out of the crowded corner than moved into it. This net movement continued until the people were evenly spread out. This models diffusion: particles moving randomly produce a net movement from a high concentration to a low concentration.

Q: (Reflection) You have now used two different models of diffusion: the picture of a slope in Figure B2.03 and moving people representing particles. Which one of these models best helped you to understand diffusion? Why?

A: Either answer is fine as long as you give a reason. The people model shows that particles move randomly, collide and change direction, and that the net movement happens by chance rather than on purpose. The slope model shows clearly which way the net movement goes (from high to low concentration) and helps you picture why a steeper gradient gives faster diffusion. Both models have limits: real particles move far faster than people, and particles do not 'roll down' a slope on purpose.

Diffusion Into and Out of Cells

- When substances diffuse into or out of a cell, they must pass through the **cell membrane** (see Question B2.02).
- **Oxygen** and **carbon dioxide** enter and leave cells by diffusion through the cell membrane, down a concentration gradient. For example, a respiring cell uses up oxygen, so there is less oxygen inside the cell than outside and oxygen diffuses in. The same cell makes carbon dioxide, so there is more carbon dioxide inside than outside and carbon dioxide diffuses out.

Factors Affecting the Rate of Diffusion

Diffusion can happen at different rates (speeds). How fast particles move from an area of high concentration to an area of low concentration depends on four factors:

Factor	What makes diffusion faster	Why
Temperature	A higher temperature	Particles have more kinetic energy, so their random movement is quicker
Surface area	A larger surface area	There is a greater area over which particles can move from one place to another
Concentration gradient	A steeper gradient (a large difference in concentration between two areas)	Particles move randomly towards the area of low concentration, and fewer move randomly in the opposite direction, so the net movement is greater
Diffusion distance	A shorter distance (e.g. a thin membrane)	Particles have less distance to travel

It is possible to investigate the effect of all four factors on the rate of diffusion in the laboratory, as the next two Experimental Skills boxes show.

Experimental Skills B2.01: Investigating How Surface Area Affects Diffusion

Aim: To use pieces of agar jelly to represent cells, and find out how the surface area of the 'cells' affects the diffusion of a substance (acid) into them.

You will need: a dish of agar jelly made with a small amount of a weak alkali (such as ammonium hydroxide) and coloured with a dye called cresol red; a sharp knife; plastic or blunt-ended forceps (tweezers); a ruler to measure in mm; two large test-tubes (more than 1 cm in diameter); a stopwatch; some dilute hydrochloric acid.

Safety: Take care with the acid and wear safety glasses in case of splashes. Clean up any spills immediately using plenty of cold water. Take care with the sharp blade: cut with the blade facing away from you, with the jelly on a firm surface.

Method:

1. Cut two cubes of agar jelly, with sides of 1 cm.
2. Cut one of the cubes into eight smaller cubes, each with sides of 0.5 cm.
3. Put the large cube into one test-tube and the eight small cubes into the other test-tube.
4. Add equal volumes of dilute hydrochloric acid to each tube, making sure all the cubes are covered with acid.
5. Start the stopwatch. Time how many seconds it takes for the cubes in each tube to become fully yellow.

How it works: cresol red is an indicator. It is red-purple in the alkaline jelly but turns yellow when the jelly becomes acidic. As the acid diffuses into a cube, the yellow colour spreads inwards from the surface. When a cube is completely yellow, the acid has reached its centre.

The numbers behind the experiment:

Measurement	One large cube	Eight small cubes
Side length	1 cm	0.5 cm
Volume of one cube	$1 \times 1 \times 1 = 1\ cm^3$	$0.5 \times 0.5 \times 0.5 = 0.125\ cm^3$
Total volume	$1\ cm^3$	$8 \times 0.125 = 1\ cm^3$
Surface area of one cube	$6 \times (1 \times 1) = 6\ cm^2$	$6 \times (0.5 \times 0.5) = 1.5\ cm^2$

Total surface area	6 cm²	8 × 1.5 = 12 cm²
Surface area to volume ratio	6 : 1	12 : 1
Distance from surface to centre	0.5 cm	0.25 cm

Expected results: the eight small cubes turn fully yellow in a shorter time than the single large cube.

Q: (1) The jelly cubes contained a weak alkali and cresol red, which is an indicator. Explain why the cubes changed colour.

A: Hydrochloric acid diffused into the jelly, from a high concentration in the acid outside the cubes to a low concentration (none) inside them. The acid neutralised the alkali in the jelly, so the jelly became acidic. Cresol red changes colour with pH, so it turned from red-purple (alkaline) to yellow (acidic). The colour change shows where the acid has diffused to.

Q: (2) How did the total volume of the eight small cubes compare with the volume of the one larger cube? Was it greater, smaller, or the same?

A: The same. One large cube has a volume of 1 × 1 × 1 = 1 cm³. Each small cube has a volume of 0.5 × 0.5 × 0.5 = 0.125 cm³, so eight small cubes have a total volume of 8 × 0.125 = 1 cm³.

Q: (3) How did the total surface area of the eight small cubes compare with the surface area of the one larger cube? Was it greater, smaller, or the same?

A: Greater (twice as big). The large cube has a surface area of 6 × (1 × 1) = 6 cm². Each small cube has a surface area of 6 × (0.5 × 0.5) = 1.5 cm², so eight small cubes have a total surface area of 8 × 1.5 = 12 cm².

Q: (4) Copy and complete these sentences, choosing the correct words: "The time taken for the eight small cubes to completely change colour was longer/shorter than the time for the one single cube. This is because the surface area/volume of the small cubes was greater/smaller than that of the single cube."

A: The time taken for the eight small cubes to completely change colour was **shorter** than the time for the one single cube. This is because the **surface area** of the small cubes was **greater** than that of the single cube.

Q: (5) Write a conclusion for your experiment. Include the words diffusion and surface area in your conclusion.

A: The larger the surface area, the faster the rate of diffusion. The eight small cubes had the same total volume as the large cube but twice the surface area (12 cm² compared with 6 cm²), so acid could diffuse into them across a larger area, and it also had a shorter distance to travel to reach the centre of each cube. This is why the small cubes changed colour completely in a shorter time.

Experimental Skills B2.02: Investigating How Temperature, Concentration Gradient or Distance Affect Diffusion

In Experimental Skills B2.01, cubes of alkaline agar jelly and an indicator were used to measure the time taken for an acid to diffuse through the cubes. In this box you **plan** how to use the same technique to investigate a different factor. The model answers below plan an investigation into **temperature**; notes for the other two factors follow at the end.

Q: (1) As well as surface area, diffusion is also affected by temperature, concentration gradient and distance. Choose one of these variables to investigate.

A: Any of the three is fine. This plan investigates temperature.

Q: (2) The variable that you change in your experiment is called the independent variable. What is the independent variable in your experiment?

A: The temperature of the hydrochloric acid that the cubes are placed in, measured in °C.

Q: (3) Think about how you can change your independent variable. Decide how many different values you will have.

A: Stand the tubes of acid in water-baths (or beakers of water) at different temperatures and check each one with a thermometer. Use at least five values to show a clear pattern, for example 10 °C (iced water), 20 °C (room temperature), 30 °C, 40 °C and 50 °C. Leave each tube of acid in its water-bath for about 10 minutes before adding the cube, so that the acid reaches the right temperature.

Q: (4) The variable that you measure, as you collect your results, is the dependent variable. What is the dependent variable in your experiment?

A: The time taken for an agar cube to turn completely yellow, measured in seconds. The shorter the time, the faster the rate of diffusion.

Q: (5) Describe how you will measure your dependent variable.

A: Start the stopwatch at the moment the cube is dropped into the acid. Watch the cube against a white background and stop the stopwatch as soon as no red-purple colour is left anywhere in the cube. Record the time in seconds.

Q: (6) It is important that no variables, other than the one you are investigating, affect your experiment. Which variables will you standardise (keep the same) in your experiment? How will you do this?

A: Keep all of these the same:

- Size of the cubes: cut every cube with 1 cm sides, measured with a ruler.
- Type of jelly: cut all the cubes from the same dish of agar, so they all contain the same amount of alkali and indicator.
- Concentration of the acid: use acid from the same bottle.
- Volume of the acid: use a measuring cylinder to put the same volume (for example 10 cm^3) into each tube, enough to cover the cube.
- Size of the tubes: use the same type of tube each time.
- End-point: the same person decides when each cube is 'fully yellow', against the same white background.

Q: (7) Write a short, clear description of how you will do your experiment.

A: A clear method could be:

1. Set up water-baths at 10, 20, 30, 40 and 50 °C.
2. Put 10 cm^3 of dilute hydrochloric acid into each of five large test-tubes and stand one tube in each water-bath for 10 minutes.
3. Cut five 1 cm cubes from the same dish of alkaline agar jelly containing cresol red.
4. Use forceps to drop one cube into the tube at 10 °C and start the stopwatch.
5. Stop the stopwatch when the cube is completely yellow and record the time.
6. Repeat for each of the other temperatures.
7. Repeat the whole experiment twice more and calculate a mean time for each temperature.

Wear safety glasses, take care with the acid and the knife, and do not heat the acid above about 50 °C.

Q: (8) Draw a results chart that you could use to fill in your results. Include headings, but do not put in any results.

A: The first column holds the independent variable and the next columns hold the dependent variable, with space for repeats and a mean. Every heading includes its unit, and units appear only in the headings:

Temperature / °C	Time to turn fully yellow, trial 1 / s	Time to turn fully yellow, trial 2 / s	Time to turn fully yellow, trial 3 / s	Mean time / s
10				

20				
30				
40				
50				

Q: (9) Predict what you would expect to find in your experiment.

A: As the temperature increases, the time taken for a cube to turn fully yellow will decrease, which means diffusion is faster at higher temperatures. This is because the acid particles have more kinetic energy at higher temperatures, so their random movement is quicker.

If you choose a different variable:

Variable chosen	Independent variable and how to change it	Prediction
Concentration gradient	The concentration of the hydrochloric acid; make up five different dilutions of the acid and keep the temperature and cube size the same	The more concentrated the acid, the steeper the concentration gradient, so the shorter the time for the cube to turn yellow
Distance	The distance the acid must travel; cut cubes of different sizes (for example sides of 0.5 cm, 1.0 cm and 1.5 cm)	The bigger the cube, the further the acid must diffuse to reach the centre, so the longer the time for the cube to turn fully yellow

B2.02 Osmosis

Water Inside and Outside Cells

Every cell in an organism's body has water inside it and outside it. Many different substances are dissolved in this water, and their concentrations may be different inside and outside the cell. This creates concentration gradients, down which water and solutes will diffuse, if they are able to pass through the membrane.

Remember: in a solution, the **solvent** is the liquid that does the dissolving (here, water) and the **solute** is the substance dissolved in it (for example, sugar).

How Osmosis Happens (Figure B2.04)

It is easiest to understand osmosis with just one solute. **Figure B2.04** shows a **dilute sugar solution** on the left separated from a **concentrated sugar solution** on the right by a membrane with very small holes, or pores, in it. The figure shows:

- Many small water molecules on both sides, with more of them on the left.
- Large sugar molecules: only a few on the left (dilute) and many on the right (concentrated).
- Arrows showing water molecules passing through the pores in both directions, but more often from left to right.
- The caption reminds you that it is important to think about the diffusion of each kind of molecule separately.

Step by step:

1. Water molecules are very small. Each one is made of two hydrogen atoms and one oxygen atom. Sugar molecules are many times larger.
2. The holes in the membrane are big enough to let water molecules through, but not sugar molecules. A membrane like this is **partially permeable**, because it lets some molecules through but not others.

3. There is a concentration gradient for sugar (higher on the right). Without the membrane, sugar molecules would diffuse into the dilute solution until they were evenly spread, but they cannot, because the holes are too small for them.
4. There is also a concentration gradient for water. On the left there is a high concentration of water molecules. On the right the concentration of water molecules is lower, because a lot of the space is taken up by sugar molecules.
5. Because there are more water molecules on the left, at any one moment more of them 'hit' a hole in the membrane and move through to the right than go the other way (right to left).
6. Over time there is an overall, or net, movement of water from left to right. This is **osmosis**.

Osmosis is really just a special kind of diffusion. It is the diffusion of water molecules in a situation where the water molecules, but not the solute molecules, can pass through a membrane.

Definition: Osmosis is the diffusion of water molecules through a partially permeable membrane.

Water Potential (Figure B2.05)

It can be confusing to talk about the 'concentration' of water molecules, because the word concentration normally means the concentration of the solute dissolved in the water. It is much better to use a different term, **water potential**:

- A **dilute solution** (where there is a lot of water) has a **high water potential**.
- A **concentrated solution** (where there is less water) has a **low water potential**.
- Pure water has the highest water potential of all.
- A difference in water potential between two areas is a **water potential gradient**. Water molecules diffuse down this gradient, from a high water potential to a low water potential.

Figure B2.05 pictures a water potential gradient as a slope. On the left, at the top of the slope, there are lots of water molecules (high water potential). A partially permeable membrane crosses the slope, and on the right, at the bottom, there are only a few water molecules (low water potential). Water molecules tend to move down the slope, from high to low water potential.

Definition (in terms of water potential): Osmosis is the net movement of water molecules from a region of higher water potential (dilute solution) to a region of lower water potential (concentrated solution) through a partially permeable membrane.

Feature	Dilute solution	Concentrated solution
Amount of solute	Little	A lot
Amount of water	A lot	Less
Water potential	High	Low
In osmosis, water moves...	Out of it, towards the concentrated solution	Into it, from the dilute solution

Water potential and osmosis are very important because they affect how and when organisms take up water. The uptake of water by plants is covered in Chapter B7, and the uptake of water by humans in Chapter B6.

Osmosis and Animal Cells (Figures B2.06 and B2.07)

Osmosis happens whenever two solutions (or a solution and pure water) are separated by a partially permeable membrane. **Cell membranes are partially permeable**: they let water through easily, but other molecules and ions often cannot get through. This means osmosis happens across cell membranes.

Animal Cell in Pure Water (Figure B2.06)

The cytoplasm inside the cell is a fairly concentrated solution. The proteins and many other substances dissolved in it are too large to get through the cell membrane, but water molecules can get through. This is just like Figure B2.04: the pure water outside is the dilute solution, the cytoplasm is the concentrated solution, and the cell membrane is the partially permeable membrane. So osmosis occurs:

1. Water molecules diffuse from the pure water (high water potential) into the cytoplasm (lower water potential), through the partially permeable cell membrane.
2. As more and more water enters, the cell swells.
3. The cell membrane has to stretch as the cell gets bigger.
4. Eventually the strain is too much and the cell **bursts**.

Figure B2.06 shows an animal cell surrounded by pure water, with an arrow showing water entering through the cell membrane.

Animal Cell in a Concentrated Solution (Figure B2.07)

If the solution outside is **more concentrated than the cytoplasm**, it has a lower water potential than the cytoplasm. Water molecules diffuse **out** of the cell through the cell membrane. As the water leaves, the cytoplasm shrinks, so the cell gets smaller. Figure B2.07 shows a shrunken, wrinkled cell with an arrow showing water leaving.

Note: if the solution outside has the same water potential as the cytoplasm, water enters and leaves at the same rate, so there is no net movement and the cell stays the same size.

Osmosis and Plant Cells in Pure Water (Figure B2.08)

Plant cells do not burst in pure water. Plant cells are surrounded by a **cell wall**, which is **fully permeable**: it lets all molecules go through it. Although it is not easy to see, a plant cell also has a **cell membrane** just inside the wall, like an animal cell, and this membrane is partially permeable.

1. A plant cell in pure water takes in water by osmosis through its partially permeable cell membrane, in the same way as an animal cell.
2. As the water goes in, the cytoplasm and vacuole swell.
3. The plant cell has a very strong cell wall, much stronger than the cell membrane. The cytoplasm presses out against the cell wall, but the wall resists and presses back on the contents.
4. This stops the plant cell from bursting.

Figure B2.08 shows a plant cell in pure water, with the cytoplasm and large vacuole pushed right up against the cell wall and an arrow showing water entering through the cell membrane.

A plant cell in this state is rather like a blown-up tyre — tight and firm. It is said to be **turgid**. The pressure of the water pushing outwards on the cell wall is called **turgor pressure**.

Why turgor matters:

- In a plant leaf, cells with plenty of water are all turgid. The pressure of the water pushing outwards on the cells' walls keeps the whole group of cells strong and firm, which helps to support the leaves.
- Turgor pressure helps a plant that has no wood in it to stay upright.
- Plant cells are usually turgid.

Plant Cells in a Concentrated Solution (Figures B2.09 and B2.10)

Now imagine a plant cell placed in a solution that is **more concentrated than its cytoplasm**:

1. Water leaves the cell by osmosis, out of the cytoplasm and vacuole, through the partially permeable cell membrane.
2. The cytoplasm and vacuole shrink.
3. The cell wall is strong and stays in position, so the cytoplasm pulls away from the cell wall, pulling the cell membrane away from the wall too.

Flaccid: when a plant cell loses a lot of water, it also loses its turgor pressure. The contents no longer push outwards on the cell wall, so the cell becomes soft and floppy. If the cells in a plant become flaccid, the plant loses its firmness and begins to **wilt**.

Plasmolysed: when the cell membrane tears away from the cell wall, the cell is said to be plasmolysed. Plasmolysis can kill a plant cell, because the cell membrane is damaged as it tears away from the cell wall.

Figure B2.09 is a diagram of a plant cell in a concentrated solution. The labels show the concentrated solution outside the cell, the less concentrated solution inside the cell, the cell wall (still the same size and shape) and the cell membrane, which has pulled away from the wall. The gap between the cell wall and the shrunken cell membrane is labelled **space X**.

Figure B2.10 is a photograph of onion cells that have been placed in a concentrated solution. The cytoplasm has lost so much water, and shrunk so much, that it has pulled away from the cell wall. The labels point to the cytoplasm, the cell wall, the cell membrane that has pulled away from the cell wall, and the space between the cell membrane and the cell wall.

Animal and Plant Cells Compared

Cell	In pure water	In a concentrated solution
Animal cell	Water enters by osmosis; the cell swells and bursts, because there is no cell wall to stop it	Water leaves by osmosis; the cytoplasm shrinks and the cell gets smaller
Plant cell	Water enters by osmosis; the cytoplasm and vacuole swell and press on the cell wall; the cell becomes turgid but does not burst	Water leaves by osmosis; the cytoplasm and vacuole shrink; the cell becomes flaccid and may become plasmolysed

Experimental Skills B2.03: Investigating the Effect of Osmosis on Potato Strips

Aim: To place pieces of plant tissue (potato) into different solutions and measure the effect on their length.

You will need: a large potato; a sharp knife or other way to peel the potato; a cork borer to cut cylinders from the potato (or use the knife to cut strips); a ruler to measure in mm; three containers such as small beakers; some distilled water, a dilute sugar solution and a concentrated sugar solution.

Safety: Take care when using the sharp knife or potato peeler.

Method:

1. Peel the potato. Use a cork borer to cut six cylinders. Cut the ends off each cylinder so that they are all the same length (40 mm is a good length) and record this length. Without a cork borer, cut rectangular pieces instead.
2. Pour the distilled water into one beaker and the two sugar solutions into the other two beakers. Label each beaker with the concentration of its solution.
3. Place two potato strips into each liquid, making sure they are completely covered.
4. Leave the strips in their liquids for about 20 minutes. While you wait, construct a results chart and fill in the original lengths.
5. After 20 minutes, take the strips out, measure the length of each one again and record the new lengths.
6. Calculate the change in length of each strip. Say whether it got longer (a positive change) or shorter (a negative change).

Figure B2.11 shows the two shapes you can cut: cylinders 40 mm long, or rectangular blocks 10 mm × 10 mm × 40 mm.

A suitable results table (headings with units; original lengths filled in before the experiment):

Liquid	Strip	Length at start / mm	Length after 20 minutes / mm	Change in length / mm
Distilled water	1	40		
Distilled water	2	40		
Dilute sugar solution	1	40		
Dilute sugar solution	2	40		
Concentrated sugar solution	1	40		
Concentrated sugar solution	2	40		

Change in length = length after 20 minutes − length at start. To compare strips that started at different lengths, you can also work out the percentage change = (change in length ÷ length at start) × 100%.

Peer assessment: swap results tables with a partner and score each feature 2 marks (really good), 1 mark (quite well done) or 0 marks (not done at all, or very poor):

- The results table is drawn with a ruler.
- There are clear headings for all the columns, with units.
- There are no units written anywhere except in the headings.
- It is really easy to understand the results table.
- The changes in length are shown in the results table.
- The changes in length have been correctly calculated, and say whether the strips increased or decreased in length.

Add up the marks you give your partner (the maximum is 12). If you have not given full marks, help your partner to see how to improve their table.

Expected results:

Liquid	Expected change in length	Why
Distilled water	Longer (positive change); the strip feels firm	Water enters the cells by osmosis and they become turgid
Dilute sugar solution	Small change, either slightly longer or slightly shorter	Depends on whether the solution has a higher or lower water potential than the potato cells
Concentrated sugar solution	Shorter (negative change); the strip feels soft and bendy	Water leaves the cells by osmosis and they become flaccid

Q: (1) What happened to the length of the potato strips that were placed in the distilled water?

A: They got longer (a positive change in length), and they also became firmer.

Q: (2) Use your knowledge of osmosis to explain why this happened.

A: Distilled water has a higher water potential than the solution inside the potato cells (their cytoplasm and vacuoles). So water moved into the cells by osmosis, through their partially permeable cell membranes, down the water potential gradient. The cytoplasm and vacuoles swelled and pushed outwards on the cell walls, so the cells became turgid. Each cell got slightly bigger, so the whole strip got longer and firmer. The cell walls stopped the cells from bursting.

Q: (3) What happened to the length of the potato strips that were placed in the concentrated solution?

A: They got shorter (a negative change in length), and they also became soft and floppy.

Q: (4) Use your knowledge of osmosis to explain why this happened.

A: The concentrated sugar solution has a lower water potential than the solution inside the potato cells. So water moved out of the cells by osmosis, through the partially permeable cell membranes, down the water potential gradient into the sugar solution. The cytoplasm and vacuoles shrank, the cells lost their turgor pressure and became flaccid (some may even have become plasmolysed), so the strip got shorter and softer.

B2.03 Active Transport

Why Cells Need Active Transport

Cells often need to take in substances that are only present in small quantities around them. If a substance has a **lower concentration outside the cell than inside**, we would expect it to diffuse **out** of the cell. If the cell needs this substance, it has to do something to make it go into the cell instead of diffusing out.

What Is Active Transport?

Cells can use energy to make substances move across their membranes **up** the concentration gradient (from a low concentration to a high concentration). This process is called **active transport**.

- The energy needed is provided by **respiration**.
- **Aerobic respiration**, in the **mitochondria** in the cell, releases energy for the cell to use. Some of this energy is used for active transport, moving substances against their concentration gradients.
- This means anything that slows down respiration (such as a shortage of oxygen) also slows down active transport.

Definition: Active transport is the movement of molecules or ions through a cell membrane from a region of lower concentration to a region of higher concentration (i.e. against a concentration gradient) using energy from respiration.

Example: Root Hair Cells Taking In Nitrate Ions

- Root hair cells in plant roots take in **nitrate ions** from the soil (plants use nitrate to make proteins).
- The concentration of nitrate ions inside a root hair cell is usually **higher** than in the soil.
- So the diffusion gradient for nitrate ions is out of the root hair and into the soil.
- But root hair cells can still take nitrate ions in, using active transport.

The bottom of this page in the book has a photograph with no caption. It appears to show root hairs: long, thin extensions growing out from the surface of a root.

How Carrier Proteins Work (Figure B2.12)

There are special **carrier proteins** in the cell membrane of root hair cells. These proteins pick up nitrate ions from outside the cell and then change shape, pushing the ions through the cell membrane and into the cytoplasm. Figure B2.12 shows a carrier protein spanning the cell membrane, with 'outside the cell' above it and 'inside the cell' below it, in three stages:

1. The nitrate ion enters the carrier protein (from outside the cell).
2. The carrier protein changes shape. The energy needed for it to do this is provided by respiration in the cell.
3. The change of shape of the carrier protein pushes the nitrate ion into the cell.

The energy for the change of shape comes from the mitochondria, which carry out aerobic respiration.

Diffusion, Osmosis and Active Transport Compared

Feature	Diffusion	Osmosis	Active transport

What moves	Any particles (molecules or ions), e.g. oxygen, carbon dioxide	Water molecules only	Molecules or ions, e.g. nitrate ions
Direction	From a higher to a lower concentration (down a concentration gradient)	From a higher to a lower water potential (dilute to concentrated solution)	From a lower to a higher concentration (against a concentration gradient)
Membrane needed?	No (it happens in air and liquids), but in cells it crosses the cell membrane	Always needs a partially permeable membrane	Happens across a cell membrane
Energy source	Kinetic energy of the particles (no energy from respiration)	Kinetic energy of the water molecules (no energy from respiration)	Energy from respiration
Carrier proteins?	No	No	Yes, they change shape to move the particles
Example in this chapter	Hydrogen sulfide spreading through a room; dialysis; acid into agar jelly	Water entering potato cells; animal cells bursting in pure water	Root hair cells taking in nitrate ions

Project B2.01: Making a Good Cup of Tea

This project comes at the end of the chapter in the book, but it is about **diffusion** and uses the planning skills from Experimental Skills B2.02.

Tea leaves are dried leaves from a camellia plant (*Camellia sinensis* is the tea plant). When hot water is added to tea leaves, colour and flavour molecules diffuse out of the leaves and into the water. Working in a group of three or four, you investigate some factors that make a really good cup of tea.

Ideas for questions to investigate (you might think of your own):

- Does the temperature of the water affect the rate at which the molecules diffuse out of the tea bag?
- Does the temperature of the water affect the flavour of the tea?
- Do different kinds of tea bag affect the diffusion of substances out of the tea leaves and into the water?
- Does adding milk to the cup before you add the water affect the diffusion of molecules out of the tea leaves?

What to do: decide on a question, then plan the experiment. Change only one variable (the independent variable), think carefully about how you will measure the dependent variable, and standardise the other variables. Predict your results, and decide how you will record them and use them to answer your question.

Example plan (water temperature):

- Independent variable: water temperature, for example 20, 40, 60 and 80 °C.
- Dependent variable: how quickly the tea colours, for example the time until a mark under the cup can no longer be seen, or comparing the colour after 2 minutes against a colour chart.
- Standardised variables: the same brand and type of tea bag, the same volume of water, the same cup, and no stirring.
- Prediction: the hotter the water, the faster the colour molecules diffuse out of the leaves, because the particles have more kinetic energy.
- Safety: take care with hot water to avoid scalds. Never taste anything made in a science laboratory; flavour tests need clean kitchen equipment.

Key Formulas and Concepts

1. Diffusion: higher concentration → lower concentration (down a concentration gradient) by random movement; the energy is the particles' kinetic energy, not respiration
2. Rate of diffusion is faster with a higher temperature, a larger surface area, a steeper concentration gradient and a shorter diffusion distance
3. Volume of a cube = side × side × side; surface area of a cube = 6 × side × side
4. One 1 cm cube: volume 1 cm^3, surface area 6 cm^2, ratio 6 : 1; eight 0.5 cm cubes: total volume 1 cm^3, total surface area 12 cm^2, ratio 12 : 1
5. Osmosis: higher water potential (dilute solution) → lower water potential (concentrated solution), through a partially permeable membrane
6. Dilute solution = a lot of water = high water potential; concentrated solution = less water = low water potential; pure water has the highest water potential
7. Animal cell: in pure water it swells and bursts; in a concentrated solution it shrinks
8. Plant cell: in pure water it becomes turgid (turgor pressure; the cell wall stops it bursting); in a concentrated solution it becomes flaccid and may become plasmolysed
9. Active transport: lower concentration → higher concentration (against a concentration gradient), through carrier proteins in the cell membrane, using energy from respiration
10. Change in length = final length − starting length (positive = longer, negative = shorter)
11. Percentage change = (change ÷ starting value) × 100%
12. Cell membrane = partially permeable; plant cell wall = fully permeable

Exam Focus Priorities

1. **Learn the full definition of diffusion** — include "net movement", "from a region of higher concentration to a region of lower concentration", "down a concentration gradient" and "as a result of random movement"; each phrase can earn credit.
2. **Diffusion and osmosis are passive** — their energy comes from the kinetic energy of the particles; never say they use energy from respiration, because only active transport does.
3. **Osmosis is only about water** — it is the diffusion of water molecules through a partially permeable membrane; the solute does not move in osmosis.
4. **Use water potential correctly** — a dilute solution has a high water potential, a concentrated solution has a low water potential, and water moves from high to low water potential.
5. **Explain why plant cells do not burst** — the strong cell wall resists the swelling contents and presses back; use the words turgid and turgor pressure, and say animal cells burst because they have no cell wall.
6. **Do not mix up flaccid and plasmolysed** — a flaccid cell has lost its turgor pressure and is soft; a plasmolysed cell has lost so much water that its cell membrane has torn away from the cell wall.
7. **The cell wall is fully permeable** — it is the cell membrane, not the cell wall, that is partially permeable, so the space between the wall and the membrane of a plasmolysed cell fills with the outside solution.
8. **Active transport answers need three ideas** — movement against a concentration gradient (low to high), carrier proteins in the membrane that change shape, and energy from respiration in mitochondria.
9. **Quote data from graphs and bar charts** — give values with units and compare them (for example "potassium is 8 times more concentrated in the cells"), then use the data to decide which process is happening.
10. **Identify anomalous results properly** — pick the point that is furthest from the line of best fit and does not fit the trend, and say why you chose it.

11. **Plan experiments using variables** — name the independent, dependent and standardised variables, repeat readings and calculate a mean, and put units only in the column headings.
12. **Show working in surface area calculations** — use cm² for area and cm³ for volume; cutting a cube into smaller cubes increases the total surface area but not the total volume.
13. **Common mistake: saying particles want to move** — particles move randomly; the net movement happens because more particles move away from the high concentration than move into it.

Textbook Questions with Answers

Question B2.01

Look at the definition of diffusion in the 'Key words' box at the start of this section. Explain why the following conditions are required for diffusion to occur:

a) movement of particles

b) a concentration gradient

c) random movement.

Answer:

a) Diffusion is the net movement of particles from one place to another, so the particles must be able to move. Their movement comes from their kinetic energy. If the particles could not move (for example, if they were held in fixed positions as in a solid), they could never spread out from a region of high concentration, so diffusion could not happen.

b) Diffusion is net movement from a region of higher concentration to a region of lower concentration, so there must be a difference in concentration. If the concentration were the same everywhere, the particles would still move about, but just as many would move one way as the other, so there would be no net movement and no diffusion.

c) The particles do not move purposefully towards the low concentration; they move randomly in all directions, changing direction when they collide. Because there are more particles in the region of high concentration, by chance more of them move out of that region than move into it. This random movement is what produces the net movement down the concentration gradient, until the particles are evenly spread out.

Question B2.02

When substances diffuse into and out of cells, which part of the cell must they move through?

Answer: They must move through the cell membrane. The cell membrane surrounds every cell, so any substance entering or leaving the cell has to cross it. (In a plant cell, substances also pass through the cell wall, but the cell wall is fully permeable, so it is the cell membrane that controls what enters and leaves.)

Question B2.03

These questions are about the diagram in Figure B2.04, which shows a dilute sugar solution (left) separated from a concentrated sugar solution (right) by a membrane with very small pores. What is the solvent in the sugar solution? What is the solute?

Answer: The solvent is water. The solute is sugar.

Question B2.04

(Figure B2.04: a dilute sugar solution and a concentrated sugar solution separated by a membrane with very small pores.) Explain why water molecules can move from one side of the membrane to the other, but sugar molecules cannot.

Answer: The membrane has very small holes (pores) in it. Water molecules are very small (each is made of just two hydrogen atoms and one oxygen atom), so they can fit through the pores. Sugar molecules are many times larger than water molecules, so they are too big to pass through the pores. The membrane is partially permeable: it lets water through but not sugar.

Question B2.05

(Figure B2.04: a dilute sugar solution on the left and a concentrated sugar solution on the right, separated by a partially permeable membrane.) In which direction is the net movement of water molecules?

Answer: From left to right, from the dilute sugar solution into the concentrated sugar solution. There are more water molecules on the left-hand side, so at any moment more of them hit a pore and pass through from left to right than pass from right to left.

Question B2.06

In Figure B2.04, which solution has the higher water potential? Explain your answer.

Answer: The dilute sugar solution (on the left) has the higher water potential. It contains more water molecules, because less of its space is taken up by sugar molecules, and a solution with a lot of water has a high water potential. The concentrated solution on the right contains less water, so it has a lower water potential.

Question B2.07

Copy and complete this sentence:

When an animal cell is placed in pure water, water …… the cell by …… through the partially permeable cell …….

Answer: When an animal cell is placed in pure water, water **enters** (moves into) the cell by **osmosis** through the partially permeable cell **membrane**.

Question B2.08

Animal cells burst if they are placed in pure water, but plant cells do not. Explain why.

Answer: Both kinds of cell take in water by osmosis in pure water, because the pure water has a higher water potential than the cytoplasm and the cell membrane is partially permeable. As water enters, the cell contents swell. An animal cell is surrounded only by its thin cell membrane, which stretches as the cell swells until the strain is too much and the cell bursts. A plant cell has a very strong cell wall outside its cell membrane. As the cytoplasm and vacuole swell, they press outwards on the cell wall, but the wall resists and presses back on the contents (turgor pressure). This stops the cell expanding any further, so the plant cell becomes turgid instead of bursting.

Question B2.09

Here are some descriptions of what happens to a plant cell when it is placed in a concentrated solution. They are in the wrong order. Write the descriptions in the correct order.

- The cytoplasm and vacuole shrink.
- The cell membrane is pulled away from the cell wall.
- Water moves out of the cell through the partially permeable cell membrane.

Answer: The correct order is:

1. Water moves out of the cell through the partially permeable cell membrane.
2. The cytoplasm and vacuole shrink.
3. The cell membrane is pulled away from the cell wall.

Water leaving the cell is what makes the cytoplasm and vacuole shrink, and because the strong cell wall stays in position, the shrinking contents pull the cell membrane away from the wall.

Question B2.10

Look at Figure B2.09, which shows a plant cell in a concentrated solution: the cytoplasm and vacuole have shrunk and the cell membrane has pulled away from the cell wall, leaving a gap labelled space X between the cell wall and the cell membrane. What fills space X? Explain your answer.

Answer: Space X is filled with the concentrated solution from outside the cell. The cell wall is fully permeable, so the outside solution (water and its dissolved solute) passes straight through it. When the cytoplasm and vacuole shrink and the cell membrane pulls away from the wall, the outside solution moves through the wall and fills the gap between the cell wall and the cell membrane. It is not cytoplasm, because the cytoplasm stays inside the cell membrane, and it is not air.

Question B2.11

A group of plant cells has been placed in a concentrated solution, and the cells are plasmolysed. Predict what will happen if the cells are now placed in pure water. Explain your answer, using the term water potential.

Answer: The cells will take up water by osmosis and become turgid again (as long as they have not been killed). Pure water has a higher water potential than the cell contents, because the cytoplasm and the solution in the vacuole are concentrated solutions with a low water potential. So water moves into the cells down the water potential gradient, through the partially permeable cell membranes. The cytoplasm and vacuole swell, the cell membrane is pushed back against the cell wall, and turgor pressure builds up, so the cells become turgid. However, if a cell membrane was badly damaged when it tore away from the cell wall, that cell may have died; its membrane would no longer be partially permeable, so it would not become turgid again.

Practice Exam Questions with Answers

Practice Question 1

Identify which change would increase the rate at which a substance diffuses across a membrane. (1 mark)

A) a decrease in concentration gradient

B) a decrease in the surface area of the membrane

C) an increase in distance across the membrane

D) an increase in temperature

Answer: D — an increase in temperature. At a higher temperature, particles have more kinetic energy, so their random movement is faster and diffusion is faster [1]. A, B and C would all slow diffusion down: a smaller concentration gradient, a smaller surface area and a longer diffusion distance all decrease the rate of diffusion.

Practice Question 2

The table shows information about diffusion, osmosis and active transport. Identify which row is correct. (1 mark)

Each option is one row of the table, giving the entries for diffusion, osmosis and active transport in that order:

A) Diffusion: uses energy from respiration. Osmosis: always involves a partially permeable membrane. Active transport: uses protein carriers in membranes.

B) Diffusion: uses kinetic energy of molecules and ions. Osmosis: uses kinetic energy of water molecules. Active transport: involves movement up a concentration gradient.

C) Diffusion: uses protein carriers in membranes. Osmosis: involves movement up a concentration gradient. Active transport: always involves a partially permeable membrane.

D) Diffusion: always involves a partially permeable membrane. Osmosis: uses protein carriers in membranes. Active transport: uses energy from respiration.

Answer: B [1]. Diffusion is powered by the kinetic energy of the moving molecules and ions; osmosis is the diffusion of water, so it is powered by the kinetic energy of the water molecules; and active transport moves substances up (against) a concentration gradient. The other rows are wrong:

- A is wrong because diffusion does not use energy from respiration.
- C is wrong because diffusion does not use protein carriers, and osmosis moves water from a high to a low water potential, not up a gradient.
- D is wrong because diffusion does not need a membrane at all (for example, a gas spreading through a room), and osmosis does not use protein carriers.

Practice Question 3

A learner investigated how temperature affected the rate at which a dye moved through a piece of agar jelly. (11 marks)

She filled seven test-tubes with a solution of red dye. She placed each tube into a water-bath at a different temperature and left them for ten minutes.

She cut seven cubes of jelly. She placed each cube into one of the tubes containing the red dye. She measured the time taken for the cubes to change colour.

This graph shows her results. It plots Time taken / minutes (y-axis, 0 to 12) against Temperature / °C (x-axis, 0 to 60). There are seven plotted points and a straight line of best fit that slopes downwards from about 12 minutes at 0 °C to below 1 minute at 60 °C. Approximate values read from the graph:

Temperature / °C	0	10	20	30	40	50	60
Time taken / minutes	12.0	10.6	4.5	6.1	4.0	2.4	1.2

a) Give the name of the process by which the dye moved through the jelly, and describe this process. [3]

b) Suggest why the learner left the tubes of red dye in the water-baths for ten minutes. [1]

c) State two variables that the learner should standardise (keep the same) in her experiment. [2]

d) i) One of the learner's results is anomalous (inconsistent). Identify this result and explain why you have chosen this value. [2]

d) ii) Describe the trend or pattern shown by the results. [1]

d) iii) Use your knowledge of diffusion to explain the results. [2]

Answer:

a) Diffusion [1]. The net movement of dye particles (molecules) from a region of higher concentration (the dye solution) to a region of lower concentration (the jelly), i.e. down a concentration gradient [1]; as a result of the random movement of the particles, which comes from their kinetic energy [1].

b) So that the dye solution in each tube had time to reach the temperature of its water-bath before the jelly cube was added, so the dye was at the temperature being tested [1].

c) Any two of [1 each, maximum 2]: the size of the jelly cubes (same side length, so the same volume and surface area); the concentration of the dye solution; the volume of dye solution in each tube; the type of jelly (all cubes cut from the same batch); how the colour change is judged (the same end-point each time).

d) i) The result at 20 °C (about 4.5 minutes) [1]. It is far below the line of best fit (the line predicts about 8 minutes at 20 °C), and it does not fit the trend: it is shorter than the time at 30 °C, even though all the other results show the time getting shorter as the temperature rises [1].

d) ii) As the temperature increases, the time taken for the cubes to change colour decreases [1] (steadily, from about 12 minutes at 0 °C to about 1 minute at 60 °C).

d) iii) At higher temperatures the dye particles have more kinetic energy, so they move faster (their random movement is quicker) [1]. So the dye diffuses into the jelly faster (the rate of diffusion increases) and the cube changes colour in less time [1].

Practice Question 4

(8 marks)

a) Describe what is meant by osmosis. [2]

b) Compare the effect of immersing an animal cell and a plant cell in pure water. Explain the reasons for any differences that you describe. [6]

Answer:

a) The net movement (diffusion) of water molecules [1] from a region of higher water potential (dilute solution) to a region of lower water potential (concentrated solution) through a partially permeable membrane [1].

b) Any six of these points [1 each, maximum 6]:

- Similarity: both cells take in water by osmosis.
- Because the pure water has a higher water potential than the cytoplasm, water moves down a water potential gradient into the cell.
- In both cells, water enters through the partially permeable cell membrane.
- In both cells, the contents swell (the cytoplasm, and in the plant cell also the vacuole).
- Difference: the animal cell keeps swelling until it bursts.
- Reason: the animal cell has no cell wall; its thin cell membrane stretches until the strain is too much and it breaks.
- Difference: the plant cell does not burst; it becomes turgid (firm).
- Reason: the plant cell has a strong cell wall; the swollen contents press outwards on the wall (turgor pressure) and the wall presses back, stopping the cell expanding further.
- The cell wall is fully permeable, so it does not stop water entering; it only stops the cell from bursting.

Practice Question 5

This bar chart shows the concentration of potassium ions and sodium ions in a sample of pond water, and in the cells of a plant growing in the water. (9 marks)

The y-axis shows Concentration / arbitrary units (0 to 4) and the x-axis shows the ion (potassium and sodium). Light bars show pond water and dark bars show plant cells. Bar heights read from the chart:

Ion	Pond water / arbitrary units	Plant cells / arbitrary units
Potassium	0.5	4.0
Sodium	0.5	about 0.05 (a very short bar, almost zero)

a) Describe the differences between the concentrations of the ions in the pond water and in the plant cells. [3]

b) Use the data in the graph to suggest how the ions move between the pond water and the plant cells. Explain your answer. [2]

c) Describe how the process you have described in your answer to b takes place. [4]

Answer:

a) Any three of [1 each, maximum 3]:

- The concentration of potassium ions is much higher in the plant cells (4.0 arbitrary units) than in the pond water (0.5 arbitrary units), about 8 times higher.
- The concentration of sodium ions is lower in the plant cells (about 0.05 arbitrary units) than in the pond water (0.5 arbitrary units).

- In the pond water, the concentrations of potassium and sodium ions are the same (0.5 arbitrary units each).
- In the plant cells, there is far more potassium than sodium (4.0 compared with about 0.05).

b) Potassium ions are moved into the plant cells by active transport [1], because they have to move from a lower concentration in the pond water to a higher concentration in the cells, which is against their concentration gradient; diffusion alone would make them move out of the cells, not in [1]. (Sodium ions are kept at a low concentration inside the cells, so they are probably actively transported out, or prevented from entering, against their concentration gradient.)

c) Any four of [1 each, maximum 4]:

- There are carrier proteins in the cell membrane.
- A potassium ion from the pond water enters (binds to) a carrier protein on the outside of the membrane.
- The carrier protein changes shape.
- The energy for the change of shape comes from respiration (aerobic respiration in the mitochondria).
- The change of shape pushes the ion through the membrane and releases it into the cytoplasm.
- This moves the ion against its concentration gradient, from a low concentration to a high concentration.

Summary of Key Concepts

Chapter Summary

- Diffusion is the net movement of molecules or ions from a high concentration to a lower concentration, as a result of their random movement and kinetic energy.
- Oxygen and carbon dioxide enter and leave cells by diffusion through the cell membranes, down a concentration gradient.
- An increase in temperature, concentration gradient or surface area speeds up diffusion into a cell. This can be investigated using pieces of agar jelly.
- Water diffuses through partially permeable membranes, including cell membranes, by osmosis.
- A solution containing a lot of water has a high water potential; in osmosis, water diffuses down a water potential gradient.
- Animal cells burst in pure water and shrink when placed in a concentrated solution. Plant cells do not burst in pure water because their cell wall stops this happening.
- A plant cell in pure water takes up water by osmosis and becomes turgid, as its contents exert outward pressure on the cell wall. A plant cell in a concentrated solution loses water by osmosis. The cell becomes flaccid and may plasmolyse.
- Active transport is the movement of particles across a cell membrane, up the concentration gradient, using energy from respiration.
- Protein carriers in cell membranes change shape to move substances across the cell membrane during active transport.

Self-Evaluation Checklist

Check that you can:

- explain what is meant by a diffusion gradient (B2.01)
- describe how to investigate factors that affect diffusion (B2.01)
- describe osmosis, referring to diffusion of water through a partially permeable membrane (B2.02)
- describe osmosis, using the term water potential (B2.02)

- describe and explain what happens when animal or plant cells are immersed in solutions of different concentrations (B2.02)
- describe active transport (B2.03)
- explain how active transport happens, including the role of protein carriers in membranes (B2.03)

Command Words in This Chapter

- **Identify:** name, select or recognise.
- **Give:** produce an answer from a given source or from recall or memory.
- **Describe:** state the points of a topic, or give its characteristics and main features.
- **Suggest:** apply your knowledge and understanding to a situation where there is a range of valid responses, in order to make proposals or put forward considerations.
- **State:** express in clear terms.
- **Explain:** set out purposes or reasons, make the relationships between things clear, say why and/or how, and support with relevant evidence.
- **Compare:** identify or comment on similarities and/or differences.

Exam Technique Tips

- Match your answer to the command word: "describe" means say what happens (with data if there is a graph); "explain" means say why it happens.
- For "compare" questions, write about both things in each sentence, for example "both cells take in water, but only the animal cell bursts".
- Count the marks: a [3] answer needs three separate, clear points.
- When a graph or chart is given, quote values with units and use comparisons such as "8 times higher".
- Always name the membrane in osmosis answers ("partially permeable cell membrane") and the energy source in active transport answers ("energy from respiration").
- In experiment questions, use the words independent variable, dependent variable and standardised variables, and describe how you would keep each standardised variable the same.